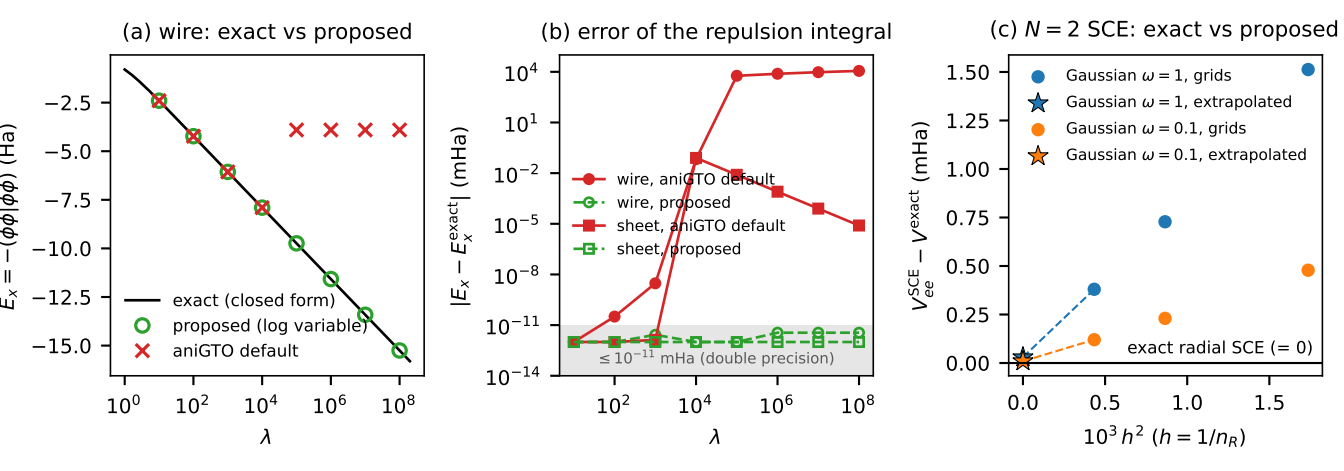
(a) wire: exact vs proposed
$E_x = -(\phi\phi|\phi\phi)$ (Ha)
−2.5
−5.0
−7.5
−10.0
−12.5
−15.0
exact (closed form)
proposed (log variable)
aniGTO default
10^0
10^2
10^4
10^6
10^8
λ
(b) error of the repulsion integral
$|E_x - E_x^{\text{exact}}|$ (mHa)
10^4
10^1
10^{-2}
10^{-5}
10^{-8}
10^{-11}
10^{-14}
wire, aniGTO default
wire, proposed
sheet, aniGTO default
sheet, proposed
$\leq 10^{-11}$ mHa (double precision)
10^2
10^4
10^6
10^8
λ
(c) $N = 2$ SCE: exact vs proposed
$V_{ee}^{\text{SCE}} - V^{\text{exact}}$ (mHa)
1.50
1.25
1.00
0.75
0.50
0.25
0.00
Gaussian $\omega = 1$, grids
Gaussian $\omega = 1$, extrapolated
Gaussian $\omega = 0.1$, grids
Gaussian $\omega = 0.1$, extrapolated
exact radial SCE (= 0)
0.0
0.5
1.0
1.5
$10^3 h^2$ ($h = 1/n_R$)